

Public Reaction to Pharmaceutical Preannouncements on Social Media: A Signaling Perspective

From Market Signals to Social Amplification of Risk

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January 13, 2026

Introduction: The Strategic Object

The Research Object: New Product Pre-Announcements (NPPA)

Definition: Strategic communications announcing a future product before market launch. (Eliashberg & Robertson, 1988).

A Strategic Communication Act: In the pharmaceutical industry, NPPAs are not just information; they are **market signals** designed to:

1. **Deterrence:** Preempt competitors by staking a claim on a therapeutic area (“First mover advantage”).
2. **Legitimacy:** Reassure stakeholders (regulators, governments) of the firm’s capability.
3. **Financial Valuation:** Reduce information asymmetry to secure investment (R&D funding is massive).
4. **Value Capture:** Prepare the market for a “Blockbuster” launch.

Core Problem: *How do these strategic signals (NPPAs), intended to create value and certainty, interact with the uncontrolled dynamics of social media to generate social amplification of risk?*

The Tension: - **Firm's Intent:** Project certainty, efficacy, control (The "Signal"). - **Social Reality:** Distortion, emotional amplification, focus on side effects (The "Noise").

The Context as a “Testing Ground”: COVID-19

Why this terrain?

The COVID-19 pandemic serves as an extreme “**stress test**” for this interaction: - **High Stakes:** Global demand vs. massive uncertainty. - **Infodemic:** An ecosystem saturated with misinformation and polarization (Pro/Anti-Vax). - **Regulatory Speed:** Rolling reviews and fast-tracks created a constant stream of “interim” signals, fueling ambiguity.

Contrast: - **1955 (Salk):** Centralized, trusted, passive reception. - **2021 (Covid):** Decentralized, skeptical, active amplification.

Chapter 1: Theoretical Framework

From Signaling to Amplification

*We move beyond the “Signal” itself to study its **Reception** and **Trajectory**.*

The Social Amplification of Risk Framework (SARF) (Kasperson et al., 1988; 2022)

Core Concept: Risk events interact with psychological, social, and institutional processes in ways that can **amplify** or **attenuate** public perceptions of risk.

Key Mechanisms: 1. **Amplification Stations:** Social media platforms (X/Twitter) act as key nodes where information is filtered and intensified. 2. **Ripple Effects:** The impact of the risk event extends beyond the direct victims (side effects) to the company (brand damage), the industry (vaccine hesitancy), and society (political unrest).

Theoretical Constructs

1. Information Mechanisms (The “What”): - **Volume:** The sheer amount of information. - **Dread Risk:** Focus on catastrophic, uncontrolled outcomes (Thrombosis vs. Headache). - **Ambiguity:** Conflicting scientific reports fuel amplification.

2. Social Mechanisms (The “Who” & “How”): - **Stigmatization:** Specific brands (AstraZeneca) become “marked” as dangerous. - **Group Polarization:** Social bubbles reinforce extreme views (Sunstein, 2002). - **Distrust:** Lack of trust in institutions acts as a multiplier for risk perception.

Hypotheses

H1: Stations of Amplification We expect to identify specific “**Amplification Stations**” (influential accounts, communities) that disproportionately drive the volume and framing of risk discourse following an NPPA.

H2: Emotional & Cognitive Response The public response to NPPAs will not be rational (Bayesian updating) but **emotionally driven**: - *Negative Bias*: Safety signals (suspensions) will trigger stronger, more durable emotional reactions than efficacy signals. - *Dread*: Discourse will focus on “dreaded” side effects (Thrombosis) rather than common ones.

H3: Signal Trajectory The social trajectory of the signal will diverge from the clinical reality: - A “brand-specific risk profile” will emerge online, constructed by social amplification rather than purely clinical data.

Methodology

Corpus: - **150,000+ Tweets** (French) mentioning Pfizer, Moderna, AstraZeneca, J&J, Sanofi, Valneva (2020-2021). - Filtered for organic user discussions (removing automated news bots).

Tools: 1. **GLiNER (Zero-Shot NER):** - Fine-tuned on a “Gold Standard” dataset (annotated with Label Studio). - Detects specific medical entities (“Thrombose”, “Myocardite”) in colloquial text. - *Why?* To measure the “Dread Risk” content precisely. 2. **Sentiment Analysis:** - XLM-RoBERTa (fine-tuned for COVID-19). - Measures the emotional valence of the amplification.

NER Pipeline (Label Studio)

Ensuring valid measurement of “Risk Signals”.

Label Studio 

Projects / Vaccine TRAIN Annotation Effets / Settings / Labeling Interface

General

Labeling Interface

Annotation

Model

Predictions

Cloud Storage

Webhooks

Danger Zone

Computer Vision >

Natural Language Processing >

Audio/Speech Processing >

Conversational AI >

Ranking & Scoring >

Structured Data Parsing >

Time Series Analysis >

Videos >

Generative AI >

Custom template

Please read the passage

The boundary of the region from which no escape is possible is called the event horizon. Although the event horizon has an enormous effect on the fate and circumstances of an object crossing it, according to general relativity it has no locally detectable features. Many steps, a black hole acts like an ideal black body, as it reflects no light.^{[2][3]} A quantum field theory in curved spacetime predicts that **black holes emit thermal radiation**; with the same spectrum as a **black body of a temperature inversely proportional to its mass**.^[4] This temperature is on the order of billions of a kelvin for black hole stellar masses, making it essentially impossible to observe directly.

Select a text span answering the following question:
How black holes could be detected?

Question Answering

Read the sentence in English

The Sun is the star at the center of the Solar System. It is a nearly perfect sphere of hot plasma, held together by its own gravity. It is the most important source of energy for life on Earth. Its diameter is about 1.39 million kilometers (864,000 miles), or 109 times that of Earth. Its mass is about 330,000 times that of Earth, and accounts for about 99.86% of the total mass of the Solar System.^[2] Roughly three quarters of the Sun's mass consists of hydrogen (75%), the rest is mostly helium (24%), with much smaller quantities of heavier elements, including oxygen, carbon, neon,

Relation Extraction

Text Classification

This is a very right on case movie that delivers everything almost right.

Choose text sentiment
☐ Positive ☐ Negative ☒ Neutral

Text Classification

Named Entity Recognition

A **Florida** restaurant paid 10,925 pounds which **Hendrix** penned on a piece of **Lo**

Named Entity Recognition

Text Summarization

Before one chooses to read the entire document in detail (2). Other examples of extraction that achieve key sequences of text in terms of abstract relevance (including categorization, summarization, and automatic/algorithm-based summarization). This has been applied mostly for text. Abstractive methods build an internal semantic representation of the original content, and then use this representation to create a summary that is closer to what a human might expect. Abstractive may transform the extracted content by paraphrasing sections of the source document, to condense a text more strongly than extraction. Such transformations, however, is computationally much more challenging than extraction, involving both natural language processing and often a deep understanding of the domain of the original text in cases where the original document relates to a specific field of knowledge. "Paraphrasing" is more difficult to apply to image and video, which is why most summarization systems are extractive.

Text Summarization

Taxonomy

To have faith is to trust yourself to the water

Opossum x Extraterrestrial

☐ Archaea
☐ Bacteria
☒ Eukarya
☐ Human
☒ Opossum
☒ Extraterrestrial

Taxonomy

Results: The Social Construction of Risk

1. The Disproportionate Focus on “Dread Risks”

Validating the SARF mechanism of “Dread”.

Top Mentioned Side Effects vs. Clinical Reality: 1. **Thrombosis** (3280 mentions) - *Rare but “Dreaded”*. 2. **Cardiac Issues** (1603 mentions) - *Rare but frightening*. 3. **Death** (474 mentions) - *The ultimate dread*. ... 6. **Fever** (180 mentions) - *Common but ignored*.

Interpretation: The “Signal” is filtered. The public amplifies risks that evoke fear (Dread), ignoring high-frequency/low-severity events.

2. Brand Stigmatization (Stigma)

Validating H3: Brand-Specific Profiles.

AstraZeneca: - Became the “Stigmatized” brand. - Associated almost exclusively with **Thrombosis** (Crisis narrative). - Volume of mentions (7,660) far exceeds Moderna (1,970) despite similar efficacy at the time.

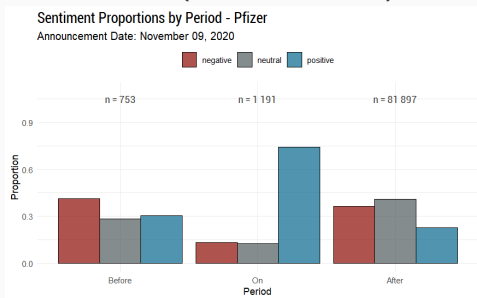
Pfizer: - The “Standard” brand. - Broadest association with diverse side effects, reflecting its ubiquity.

Moderna: - The “Ghost” brand. - Less distinct narrative, “Me-too” effect.

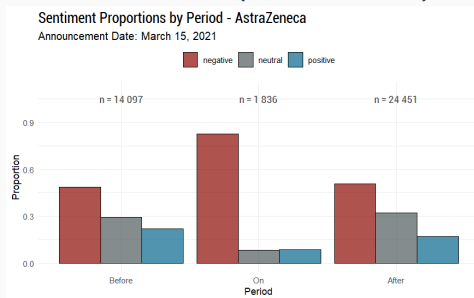
3. Asymmetry of Amplification (Ripple Effects)

Validating H2: Negative Bias.

Pfizer (Efficacy Signal)



AstraZeneca (Safety Signal)



- **Positive Signal (Left):** Spike is sharp but **transient**. Optimism decays rapidly.
- **Negative Signal (Right):** Spike is massive and **persistent**. The negative sentiment creates a new “baseline” (Stigma).

Discussion & Conclusion

1. **Digital SARF:** We empirically demonstrate SARF in a digital context.
 - *Stations:* Twitter acts as a high-speed amplification station.
 - *Mechanisms:* Dread Risk and Stigmatization are the primary drivers of the narrative.
2. **The Limits of Signaling:**
 - Traditional Signaling Theory assumes a rational receiver (Market).
 - We show that in a social context, the receiver is **emotional** and **constructivist**.
 - The “Signal” (NPPA) is just the trigger; the “Meaning” is constructed by the crowd.

Managerial Implications

- **Managing the “Ripple Effects”:** Firms cannot just “broadcast” positive data. They must actively monitor and manage the **secondary ripple effects** (stigmatization).
- **Addressing Dread:** Communication must address *dread risks* (even if rare) with extreme transparency, as these drive the amplification loop.
- **The “Silence” Danger:** Leaving the narrative void after a positive announcement allows negative amplification to take over (as seen in the rapid decay of optimism).

Thank you for your attention.